

Long Nguyen Khac

📍 Hanoi, Vietnam | ✉ nk.long723@gmail.com | 🏠 longnk0702.github.io | 📷 longnk0702 | 🌐 longnk0702 | 🎓 Google Scholar

Education

Vietnam National University, Hanoi – International School

Hanoi, Vietnam

B.Eng. in Automation and Informatics

Sep. 2021 – Dec. 2026

- **GPA:** 3.71/4.00; ranked in the top 4% of the cohort.
- Completed a one-year gap period for international research and exchange programs in Indonesia, Taiwan and Singapore (05/2025 – 05/2026).
- **Research interest:** robotics, wireless power transfer, and AI.

Research Experience

Water & Energy Research Lab, School of EEE, Nanyang Technological University

Singapore

Research Fellow

Feb. 2026 – Mar. 2026

- Conducted research on relay-coil-enhanced spatial wireless power transfer systems for autonomous underwater vehicle charging.
- Analyzed coil configurations, power transfer efficiency, and system-level integration for docking-free underwater charging.
- Explored mechanical and communication requirements for integrating wireless charging into an AUV prototype.

AISC Lab, AIM-HI Institute, National Chung Cheng University

Chiayi, Taiwan / Remote

Research Intern

May 2025 – Present

- Applying machine learning models, including Random Forest, XGBoost, and Support Vector Regression, to predict cutting forces in CNC milling from Taguchi machining datasets.
- Developed data-driven modeling approaches for cutting force prediction in CNC milling using experimental datasets.
- Applied nonlinear system analysis, sparse identification, and machine learning methods to model machining dynamics.
- Built data processing and model evaluation pipelines for machining force prediction under different experimental conditions.

Intelligent Control and Artificial Intelligence Laboratory, VNUIS

Hanoi, Vietnam

Student Researcher

Sep. 2022 – Present

- Conducting research on adaptive control for AUVs, focusing on station keeping for planar wireless charging.
- Designed the mechanical model and physical prototype of a 3-DOF delta parallel robot, and implemented adaptive fuzzy backstepping control for robust trajectory tracking under uncertainties.
- Built a dataset of 2,632 construction-site images and trained YOLOv5s-based models for personal protective equipment detection.

Publications

1. **Long Nguyen Khac** et al. *Adaptive Fuzzy Logic in Backstepping Control for 3-DOF Parallel Robot*. International Conference on Advances in Information and Communication Technology, Springer, 2024. DOI: 10.1007/978-3-031-80943-9_57.
 - Main highlights: Adaptive fuzzy backstepping control for robust trajectory tracking of a 3-DOF delta parallel robot.
 - Presentation: Mechatronics and Automation section, The 3rd International Conference ICTA 2024.
2. **Long Nguyen Khac** et al. *YOLOv5s Model with Applied Activation Functions for Personal Protective Equipment Detection in Construction Sites*. Proceedings of the National Conference on Smart Technology Applications in Industry 4.0, Smart City, and Sustainability, 2023.
 - Main highlights: YOLOv5s-based PPE detection using six activation functions and a dataset of 2,632 construction-site images.
 - Presentation: National Conference on Smart Technology Applications in Industry 4.0, Smart City, and Sustainability, 2023.

Selected Honors & Awards

Fully Funded Scholarship, NUS Enterprise Summer Programme in Entrepreneurship, NUS ASEAN SPARK Fellowship	Jul. 2026
Most Popular Poster Award, NTU Singapore Global Connect Fellowship Final Research Poster Presentation	Mar. 2026
Exchange Scholarship, Kyushu University Spring-Semester Exchange Scholarship, Department of Mechanical Engineering	Jan. 2026
Research Internship Scholarship, Taiwan Experience Education Program, Ministry of Education, Taiwan	May. 2025
Scholarship, TOSHIBA Scholarship, TOSHIBA International Foundation	Sep. 2024
Fully Funded Exchange Program, Temasek Foundation Specialist's Community Action & Leadership Exchange	Feb. 2024
Fighting Spirit Prize, South-East Asia Circuit and System Society Hackathon	Dec. 2022
Merit-based University Academic Scholarship, Six-semester recipient for excellent academic performance	2021–2025

Selected International Experience

School of Electrical and Electronic Engineering, Nanyang Technological University

Singapore

Global Connect Fellow

Feb. 2026 – Mar. 2026

- Selected as 1 of 39 young scholars worldwide and 1 of 5 outstanding Vietnamese representatives for a fully funded fellowship at NTU, Singapore (~2% acceptance rate from ~2,700 applicants).
- Joined an international research environment with exposure to interdisciplinary work across energy systems, and robotics.

College of Design and Engineering, National University of Singapore

Singapore

Exchange Student

Aug. 2025 – Dec. 2025

- Selected as one of 8 outstanding VNU students receiving a fully funded scholarship for one semester exchange in NUS.
- Completed a one-semester academic exchange, gaining broader exposure to engineering education, international research culture, and interdisciplinary learning.

College of Engineering, National Chung Cheng University

Taiwan

Research Intern

May 2025 – Jul. 2025

- Selected as one of 28 global students to intern at National Chung Cheng University funded by Taiwan Experience Education Program (TEEP), Ministry of Education, Taiwan
- Worked at Advanced Institute of Manufacturing with High-tech Innovations (AIM-HI) – one of leading tech institutions in Taiwan, participating in smart sensing manufacturing research project.

- Selected as one of four excellent VNU representatives to participate the program funded by Kyushu University, Japan.
- Participated in an ASEAN-focused academic program, strengthening regional understanding, cross-cultural communication, and awareness of science, technology, and society in Southeast Asia.

Temasek Foundation Specialist's Community Action & Leadership Exchange

- Selected as one of 29 VNU students to receive a scholarship for the program.
- Contributed to a group project on youth leadership and sustainability in urban development in Vietnam and Singapore.
- Engaged in a regional leadership and community action program, developing teamwork, communication, and social responsibility in an international environment.

Selected Teaching Experience

Faculty of Engineering and Technology, VNUIS

- Modeling and Simulation of Control Systems**
 - Assisted 30 students across 7 laboratory sessions on MATLAB/Simulink for robotics, mechatronics, and process control applications.
 - Supported the preparation of laboratory manuals, tutorial exercises, and simulation-based learning materials.
- Microprocessors and Microcontrollers**
 - Mentored 30 students through 15 laboratory sessions on Embedded C, Arduino, STM32, and embedded system development.

Selected Working Experience

ETEK Automation Solutions JSC

- Project:** AGV Virtual Simulation for Automated Warehouse Logistics – VINFAST Hai Phong.
 - Sketched AGV/AMR traffic intersections using AutoCAD and simulated warehouse logistics operations using Visual Components Premium 4.6.
 - Monitored AGV/AMR behavior, identified simulation errors, and supported troubleshooting of automated logistics workflows.

Vietnam E-Commerce and Digital Economy Agency, Ministry of Industry and Trade

- Project:** Consultancy Support and Sales Policy Development for E-Commerce Website.
 - Coordinated with SMEs across Dak Nong and An Giang provinces to support e-commerce platform development.
 - Assisted in website design, content management, and preparation of technical manual documents.

Skills

Research:	Mathematical modeling, control systems design, autonomous systems (AUV, mobile robots), wireless power transfer, data-driven modeling
Programming:	MATLAB, Python, Embedded C/C++, PLC programming
Softwares:	MATLAB/Simulink, SolidWorks, Altium, Proteus, Visual Components, TIA Portal
AI:	Data-driven modeling (SINDy, SVD), Machine learning (Random Forest, XGBoost), Computer vision (YOLO)
Languages:	Vietnamese (Native), English (Fluent), French (Beginner), Chinese (Beginner)

References

Asst. Prof. Yun Yang Assistant Professor, School of Electrical & Electronic Engineering, NTU

- Website: Yun Yang
- Contact: yun.yang@ntu.edu.sg

Prof. Her-Terng Yau

Director of AIM-HI, National Chung Cheng University

- Website: Her-Terng Yau
- Contact: htyau@ccu.edu.tw

Assoc. Prof. Nguyen Nhu Tung

Senior Lecturer, Faculty of Engineering and Technology, VNUIS

- Website: Nhu-Tung Nguyen
- Contact: tungnn@vnu.edu.vn

Dr. Le Xuan Hai

Vice Dean, Faculty of Engineering and Technology, VNUIS

- Website: Hai-Xuan Le
- Contact: hailx@vnu.edu.vn